

1.输入输出，类的入门

C++

```
1  /*#include<stdio.h>
2  int main()
3  {int a,b;
4      scanf("%d%d",&a,&b);
5      printf("%d",a+b);}
6  */
7  #include<iostream>
8  using namespace std;
9  int main()
10 {double a,b;
11     int x=999;
12     cin>>a>>b;//输入
13     cout<<a+b<<endl;//输出 相加
14     cout<<a<<"+"<<b<<"="<<a+b<<' '<<x<<endl;//end of line}
```

C++

```
1  //要求：使用switch语句，使用cout
2  //输入输出示例：
3  //输入1: 1
4  //输出1: Monday
5  //输入2: 7
6  //输出2: Sunday
7  #include<iostream>
8  using namespace std;
9  int main(){
10     int day;
11     cin>>day;
12     switch(day){
13         case 1:
14             cout<<"Monday"<<endl;
15             break;
16         case 2:
17             cout<<"Tuesday"<<endl;
18             break;
19         case 3:
20             cout<<"Wednesday"<<endl;
21             break;
22         case 4:
23             cout<<"Thursday"<<endl;
24             break;
25         case 5:
26             cout<<"Friday"<<endl;
27             break;
28         case 6:
29             cout<<"Saturday"<<endl;
30             break;
31         case 7:
32             cout<<"Sunday"<<endl;
33             break;
34         default:
35             cout<<"error"<<endl;
36             break; }
37     return 0;
38 }
```

C++

```
1  #include <iostream>
2  using namespace std;
3  class fractionAdd{
4  private:
5      int numerator;
6      int denominator;
7      int gcd(int a,int b){
8          while(b≠0)
9              {int r=a%b;
10                 a=b;
11                 b=r;}
12          return a;//辗转相除求最大公因数
13      }
14  public:
15      fractionAdd(int n1,int n2,int n3,int n4){
```

```

16     numerator=n1*n4+n2*n3;
17     denominator=n2*n4;
18     int divisor=gcd(numerator,denominator);
19     numerator/=divisor;
20     denominator/=divisor;}
21     void Show(){cout<<numerator<<"/"<<denominator<<endl;}
22 };
23 int main(){
24     int a,b,c,d;
25     cin>>a>>b>>c>>d;
26     fractionAdd frac(a,b,c,d);
27     frac.Show();
28     return 0;}

```

C++

```

1  #include<iostream>
2  using namespace std;
3  class Fraction{
4  private:
5      int numerator;
6      int denominator;
7      int gcd(int a,int b)
8      {while(b!=0)
9      {int r=a%b;
10     a=b;
11     b=r;
12     }
13     return a;
14     }
15 public:
16     Fraction(int n,int d)
17     {
18         int divisor=gcd(n,d);
19         numerator=n/divisor;
20         denominator=d/divisor;
21     }
22     Fraction(int n)
23     {
24         numerator=n;
25         denominator=1;
26     }
27     Fraction()
28     {
29         numerator=1;
30         denominator=1; // 重载
31     }
32     void show()
33     {
34         cout<<numerator<<"/"<<denominator;
35     }
36 };
37 //StudybarCommentBegin
38 int main()
39 {
40     Fraction f1,f2(2),f3(1,3);
41     f1.show(); cout<<endl;
42     f2.show(); cout<<endl;
43     f3.show(); cout<<endl;
44 }
45 //StudybarCommentEnd

```

时间类

C++

```

1  #include<iostream> //基本时间类
2  using namespace std;
3  class Cmytime{
4  private:
5      int h,m,s;
6  public:
7      void Set(int hh,int mm,int ss){h=hh;m=mm;s=ss;};
8      void Show(){cout<<h<<":"<<m<<":"<<s<<endl;};
9  };
10 //StudybarCommentBegin
11 int main(void) {
12     int h,m,s;
13     cin>>h>>m>>s;
14     Cmytime t1;//t1是对象，是零件，object
15     t1.Set(h,m,s);
16     t1.Show();
17     return 0;
18 }
19 //StudybarCommentEnd

```

```

1  #include<iostream>
2  using namespace std;
3  class Cmytime{
4  private:
5      int h,m,s;
6  public:
7      Cmytime()
8      {}
9      Cmytime(int hh,int mm,int ss)
10     {
11         h=hh;
12         m=mm;
13         s=ss;
14     }
15     void Set(int hh,int mm,int ss){h=hh;m=mm;s=ss;};
16     void Show(){cout<<h<<":"<<m<<":"<<s<<endl;};
17 };
18 //StudybarCommentBegin
19 int main(void) {
20     int h,m,s;
21     cin>>h>>m>>s;
22     Cmytime t1(3,2,1);
23     t1.Show();
24     cout<<"\n";
25     // Cmytime t3;
26     // t3.Show();
27     // cout<<"\n";
28     t1.Set(h,m,s);
29     t1.Show();
30     return 0;
31 }
32 //StudybarCommentEnd

```

```

1  #include<iostream>
2  using namespace std;
3  class Cmytime{
4  private:
5      int h,m,s;
6  public:
7      int Set(int hh,int mm,int ss){
8          if (0<=hh&&hh<=23&&0<=mm&&mm<=59&&0<=ss&&ss<=59){
9              h=hh;
10             m=mm;
11             s=ss;
12             return 1;}
13         else{
14             // cout<<"invalid value"<<endl;
15             return 0;}
16         }
17
18     void AddOneSecond(){//时间加一秒
19         s=s+1;
20         if(s>60){
21             s=s-60;
22             m=m+1;
23         }
24         if(m>60){
25             m=m-60;
26             h=h+1;
27         }
28         if(h>23)
29         {
30             h=h-24;
31         }
32     }
33     void Show(){
34         cout<<h<<":"<<m<<":"<<s<<endl;
35     }
36 };
37
38 //StudybarCommentBegin
39 int main(void) {
40     int h,m,s;
41     cin>>h>>m>>s;
42
43     Cmytime t1;
44     t1.Set(h,m,s);
45     t1.Show();
46     cout<<endl<<t1.Set(24,0,0)<<"\n";
47     t1.Show();
48

```

```
49     t1.AddOneSecond();
50     cout<<endl;
51     t1.Show();
52
53     return 0;
54 }
55 //StudybarCommentEnd
```

C++

```
1  #include<iostream>
2  using namespace std;
3  class Cmytime{
4  private:
5      int h,m,s;
6  public:
7      int Set(int hh,int mm,int ss){
8          if (0<=hh&&hh<=23&&0<=mm&&mm<=59&&0<=ss&&ss<=59){
9              h=hh;
10             m=mm;
11             s=ss;
12             return 1;}
13         else{
14             // cout<<"invalid value"<<endl;
15             return 0;}
16         }
17
18         int AddNSeconds(int n) { //时间加N秒
19             s = s + m * 60 + h * 3600 + n;
20             h = s / 3600;
21             m = (s % 3600) / 60;
22             s = s % 60;
23             int day=h/24;
24             return day;
25         }
26         void Show() {
27             cout << h << ":" << m << ":" << s;
28         }
29 };
30
31 //StudybarCommentBegin
32 int main(void) {
33     int h,m,s;
34     cin>>h>>m>>s;
35     Cmytime t1;
36     t1.Set(h,m,s);
37     t1.Show();
38     cout<<endl<<t1.Set(24,0,0)<<"\n";
39     t1.Show();
40     t1.AddNSeconds(1);
41     cout<<endl;
42     t1.Show();
43     cout<<endl<<t1.AddNSeconds(3600*25);
44     return 0;
45 } //StudybarCommentEnd
```

C++

```
1  #include<iostream>
2  #include<iomanip>
3  #include <cmath>
4  using namespace std;
5  class Cmycomplex{
6  private:
7      double x,y;
8  public:
9      Cmycomplex(double xx=0,double yy=0){x=xx;y=yy;}
10     void Set(double xx,double yy)
11     {
12         x=xx;y=yy;
13     }
14     Cmycomplex operator+(const Cmycomplex&z) //返回值类型是Cmycomplex
15     {
16         Cmycomplex t;
17         t.x=x+z.x;
18         t.y=y+z.y;
19         return t;
20     }
21     Cmycomplex operator-(const Cmycomplex&z) //返回值类型是Cmycomplex
22     {
23         Cmycomplex t;
24         t.x=x-z.x;
25         t.y=y-z.y;
26         return t;
27     }
28     Cmycomplex operator*(const Cmycomplex&z) //返回值类型是Cmycomplex
```

```

29     {
30         Cmycomplex t;
31         t.x=x*z.x-y*z.y;
32         t.y=x*z.y+z.x*y;
33         return t;
34     }
35     Cmycomplex operator/(const Cmycomplex&z) //返回值类型是Cmycomplex
36     {
37         Cmycomplex t;
38         t.x=(x*z.x+y*z.y)/(z.x*z.x+z.y*z.y);
39         t.y=(z.x*y-x*z.y)/(z.x*z.x+z.y*z.y);
40         return t;
41     }
42
43     Cmycomplex sqrt() //返回值类型是Cmycomplex
44     {
45         if(x<0 && y==0)
46         {
47             Cmycomplex t;
48             t.x=0;
49             t.y::sqrt(-x);
50             return t;
51         }
52         if (y == 0 && x == 0) {
53             return Cmycomplex(0, 0);
54         }
55         Cmycomplex t;
56         t.x::sqrt((x::sqrt(x*x+y*y))/2);
57         t.y=y/::sqrt(2*(x::sqrt(x*x+y*y)));
58         return t;
59     }
60     void Show(){
61         // 具体显示几位小数的设置
62
63         std::cout<<"("<<fixed<<setprecision(2)<<x; // 声明空间
64         if (y>=0){
65             std::cout<< "+"<<fixed<<setprecision(2)<<y<< "i)"<<endl;
66         }
67         if (y<0){
68             std::cout<<fixed<<setprecision(2)<<y<< "i)"<<endl;
69         }
70         double gety()const{
71             return y;
72         }
73         friend Cmycomplex operator*(const double&n,const Cmycomplex&z);
74         friend Cmycomplex operator/(const double&n,const Cmycomplex&z); //返回值类型是Cmycomplex
75     };
76     Cmycomplex operator*(const double&n, const Cmycomplex&z){//友元函数
77         Cmycomplex t;
78         t.x=n*z.x;
79         t.y=n*z.y;
80         return t;
81     }
82     Cmycomplex operator/(const double&n, const Cmycomplex&z){
83         Cmycomplex t;
84         t.x=z.x/n;
85         t.y=z.y/n;
86         return t;
87     }
88
89     int main()
90     {
91         double x1, x2, x3, y1, y2, y3;
92         cin >> x1 >> y1;
93         cin >> x2 >> y2;
94         cin >> x3 >> y3;
95         Cmycomplex a(x1, y1), b(x2, y2), c(x3, y3), z, t1, t2;
96         z=b*b-4*a*c;
97         t1=((-1)*b+z.sqrt())/2*a; //z.sqrt()为求复数z的平方根，这里的2*a涉及操作符重载，且友元重载。
98         t2=((-1)*b-z.sqrt())/2.0/a; //这里涉及到除法的重载
99         if(t1.gety(>t2.gety() //gety()为得到虚部值
100     {
101         t1.Show();
102         t2.Show();
103     }
104     else
105     {
106         t2.Show();
107         t1.Show();
108     }
109     return 0;
110 }

```